

Scalability & Future Plan

Date	15 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Uses a synthetic/small dataset instead of a real banking dataset	Prediction accuracy may not represent real-world scenarios	High
2	No user authentication and database storage	Prediction history and user management are unavailable	Medium
3	Single-server deployment on Render Free Tier	Limited ability to handle high traffic and concurrent users	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a small number of users simultaneously	Deploy using AWS, Azure, or Kubernetes with load balancing
2	Data Storage	No database integration	Integrate MySQL or MongoDB to store user and prediction data
3	Performance	Predictions are performed on a single server	Use model caching and cloud auto-scaling services
4	Security	Basic Flask application without authentication	Add Login System, JWT Authentication, and HTTPS encryption

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate real-world banking dataset and retrain models	Within 2 months	Improves prediction accuracy and reliability
Phase 3	Add user authentication and prediction history database	Within 3 months	Better user experience and data management
Phase 4	Deploy scalable cloud infrastructure and develop a mobile application	Within 6 months	Supports large-scale users and improves accessibility